

Azure SQL Database Health Check — Key Steps for Optimal Performance

When performing a **health check for Azure SQL Database**, the goal is to ensure **availability, performance, security, and cost optimization**. Here's the structured checklist:

1 Check Database Availability

- Confirm all Azure SQL databases and elastic pools are in the **"Online"** state.
- Use **Azure Portal**, **PowerShell**, or **Azure CLI** to verify connectivity.
- Test connectivity via SQL Server Management Studio (SSMS) or Azure Data Studio.

2 Review Resource Utilization

- Monitor key metrics in **Azure Monitor** or **Metrics Explorer**:
 - **DTU/CPU utilization**
 - **IO latency (Read/Write)**
 - **Data and log I/O percentage**
 - **Memory and tempdb usage**
- Ensure the service tier matches workload demands (avoid over/under-provisioning).

3 Validate Performance Insights

- Open **Query Performance Insight** in the Azure Portal.
- Identify top resource-consuming queries (CPU, IO, Duration).
- Tune indexes or queries where necessary.
- Use **Query Store** to analyze query regressions or plan changes.

4 Check Backups and Restore Options

- Confirm **automatic backups** are running (Azure performs daily backups automatically).
- Validate **point-in-time restore** availability.
- For business-critical workloads, verify **long-term backup retention (LTR)** policies.

5 Verify High Availability & DR Readiness

- Review **geo-replication** status (if configured).
- Ensure the **failover group** is synchronized and automatic failover works.
- Test read-access geo-secondary connections.

6 Review Security & Compliance

- Ensure **Azure Defender for SQL** is enabled.
- Review **auditing and threat detection** logs.
- Check encryption settings:
 - **Transparent Data Encryption (TDE)**
 - **Always Encrypted** (for sensitive columns)
 - **Firewall rules / Private Endpoints**
- Validate least-privilege access via **Azure Active Directory authentication**.

7 Inspect Index and Statistics Health

- Check for **index fragmentation** and stale statistics.
- Run **Azure SQL Maintenance scripts** or use **Automated Tuning** options for:
 - Automatic index creation
 - Automatic index drop
 - Automatic statistics updates

8 Validate Alerts and Monitoring

- Confirm **alerts** are configured for CPU, DTU, storage, and deadlocks.
- Ensure **Azure Log Analytics** is collecting performance and diagnostic logs.
- Review insights in **Azure Monitor Workbooks** or **Power BI dashboards**.

9 Review Cost and Scaling

- Check **pricing tier** and usage patterns.
- Use **Azure Advisor** recommendations for cost optimization.
- Scale up/down or switch between **DTU-based** and **vCore-based** models as needed.

10 Review Configuration & Connectivity

- Verify **connection resiliency** (retry logic in applications).
- Confirm **maintenance window** settings align with business needs.
- Check **COLLATION**, **MAXDOP**, and **Query Store settings**.

✓ Final Validation

- Review last week's alerts, incidents, or performance anomalies.
- Export metrics and logs for documentation and audit.
- Ensure proactive monitoring and periodic reviews are scheduled.